Level 1

Python Basics & Data Types

Variables, Data Types, Strings, Numbers, Booleans, Input/Output

Question1: Print “Hello, Hacker!” to the console

```
print("Hello, Hacker!")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 1.py
Hello, Hacker!
```

Question2: Ask the user for their name and print "Welcome, [name]!".

```
name = input("What is your name? ")
print(f"Welcome, {name}!")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 2.py
What is your name? NadaAbuzaid
Welcome, NadaAbuzaid!
```

Question3: Convert "1337" (string) to an integer and add 10.

```
number = int("1337")
result = number + 10
print(f"The result is: {result}")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 3.py
The result is: 1347
```

Question4: Create a program that asks for two numbers and prints their sum, difference, product, and quotient.

```
num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))

total = num1 + num2
difference = num1 - num2
product = num1 * num2
quotient = num1 / num2 if num2 != 0 else "undefined"

print(f"Sum: {total}, Difference: {difference}, Product:
{product}, Quotient: {quotient}")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 4.py
Enter the first number: 15
Enter the second number: 20
Sum: 35, Difference: -5, Product: 300, Quotient: 0.75
```

Question5: Reverse the string "rekcah_repus" without using built-in functions.

```
# Solution 1:
print("rekcah_repus"[::-1])

# Solution 2:
text = "rekcah_repus"
reversed_text = ""

for char in text:
    reversed_text = char + reversed_text

print(reversed_text)
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 5.py
super_hacker
```

Question6: Write a program that checks if a number is even or odd.

```
num  = int(input("Enter a number: "))
if num % 2 == 0:
    print(f"{num} is even")
else:
    print(f"{num} is odd")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 6.py
Enter a number: 5
5 is odd
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 6.py
Enter a number: 8
8 is even
```

Question7: Create a variable that stores a boolean value and print it.

```
is_true = True
print(f"The boolean value is: {is_true}")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 7.py
The boolean value is: True
```

Question8: Convert the string "100101" to a decimal number.

```
string_num = "100101"
print(f"The decimal value of {string_num} is: {int(string_num,
2)}")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 7.py
The boolean value is: True
```

Question9: Print "H4ck3r" with alternating uppercase and lowercase letters.

```
text = "H4ck3r"
count = 0

for char in text:
    if char.isalpha():
        if count % 2 == 0:
            print(char.lower(), end="")
        else:
            print(char.upper(), end="")
        count += 1
    else:
        print(char, end="")
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 9.py
h4Ck3R%
```

Question10: Replace all vowels in "P@ssw0rd" with "*".

```
text = "P@ssw0rd"
result = ""

for char in text:
    if char.lower() in "aeiou":
        result += "*"
    else:
        result += char

print(result)
```

```
(base) nadaabuzaid@Nadas-MacBook Level1 % python3 10.py
P@ssw0rd
```